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Circulation

ORIGINAL RESEARCH ARTICLE

Circulating MicroRNA-122-5p Is Associated With a Lack of Improvement in Left Ventricular Function After Transcatheter Aortic Valve Replacement and Regulates Viability of Cardiomyocytes Through Extracellular Vesicles

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BACKGROUND: Transcatheter aortic valve replacement (TAVR) is a well-established treatment option for high- and intermediate-risk patients with severe symptomatic aortic valve stenosis. A majority of patients exhibit improvements in left ventricular ejection fraction (LVEF) after TAVR in response to TAVR-associated afterload reduction. However, a specific role for circulating microRNAs (miRNAs) in the improvement of cardiac function for patients after TAVR has not yet been investigated. Here, we profiled the differential expression of miRNAs in circulating extracellular vesicles (EVs) in patients after TAVR and, in particular, the novel role of circulating miR-122-5p in cardiomyocytes.

METHODS: Circulating EV-associated miRNAs were investigated by use of an unbiased Taqman-based human miRNA array. Several EV miRNAs (miR-122-5p, miR-26a, miR-192, miR-483-5p, miR-720, miR-885-5p, and miR-1274) were significantly deregulated in patients with aortic valve stenosis at day 7 after TAVR compared with the preprocedural levels in patients without LVEF improvement. The higher levels of miR-122-5p were negatively correlated with LVEF improvement at both day 7 (r=–0.264 and P=0.015) and 6 months (r=–0.328 and P=0.0018) after TAVR.

RESULTS: Using of patient-derived samples and a murine aortic valve stenosis model, we observed that the expression of miR-122-5p correlates negatively with cardiac function, which is associated with LVEF. Mice with graded wire injury–induced aortic valve stenosis demonstrated a higher level of miR-122-5p, which was related to cardiomyocyte dysfunction. Murine ex vivo experiments revealed that miR-122-5p is highly enriched in endothelial cells compared with cardiomyocytes. Coculture experiments, copy-number analysis, and fluorescence microscopy with Cy3-labeled miR-122-5p demonstrated that miR-122-5p can be shuttled through large EVs from endothelial cells into cardiomyocytes. Gain- and loss-of-function experiments suggested that EV-mediated shuttling of miR-122-5p increases the level of miR-122-5p in recipient cardiomyocytes. Mechanistically, mass spectrometry, miRNA pulldown, electrophoretic mobility shift assay, and RNA immunoprecipitation experiments confirmed that miR-122-5p interacts with the RNA-binding protein hnRNPU (heterogeneous nuclear ribonucleoprotein U) in a sequence-specific manner to encapsulate miR-122-5p into large EVs. On shuttling, miR-122-5p reduces the expression of the antiapoptotic gene *BCL2* by binding to its 3′ untranslated region to inhibit its translation, thereby decreasing the viability of target cardiomyocytes.

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CONCLUSIONS: Increased levels of circulating proapoptotic EV-incorporated miR-122-5p are associated with reduced LVEF after TAVR. EV shuttling of miR-122-5p regulates the viability and apoptosis of cardiomyocytes in a BCL2-dependent manner.

Key Words: cardiovascular diseases ■ endothelial cells ■ extracellular vesicles ■ microRNAs ■ myocytes, cardiac ■ transcatheter aortic valve replacement

Clinical Perspective

What Is New?

- Aortic valve stenosis increases circulating microRNA-122-5p (miR-122-5p), which correlates with a lack of improvement of the left ventricular ejection fraction in patients after transcatheter aortic valve replacement.
- Extracellular vesicles harbor miR-122-5p and facilitate its shuttling into cardiomyocytes. Vesicular shuttling of miR-122-5p is regulated by a direct interaction with a multifunctional RNA-binding protein, hnRNPU (heterogeneous nuclear ribonucleoprotein U), in a sequence-specific manner.
- Extracellular vesicles containing miR-122-5p post-transcriptionally repress *BCL2*, an antiapoptotic gene, which is central to cell viability and apoptosis.

What Are the Clinical Implications?

- The levels of circulating and extracellular vesicular miR-122-5p were found to be an indicator of heart function in patients with low or no left ventricular ejection fraction improvement after transcatheter aortic valve replacement.
- An increase in extracellular vesicular miR-122-5p in patients with aortic valve stenosis represents a novel mechanism for the deterioration of cardiac function in patients after transcatheter aortic valve replacement.
- Pharmacological manipulation of extracellular vesicle–incorporated miR-122-5p levels may improve left ventricular ejection fraction and cardiac function in patients with aortic valve stenosis by improving the viability of cardiomyocytes, which provides a therapeutic possibility for patients with limited left ventricular ejection fraction improvement.

Nonstandard Abbreviations and Acronyms

ACS	acute coronary syndrome
AVS	aortic valve stenosis
EC	endothelial cell
EV	extracellular vesicle
hnRNP	heterogeneous nuclear ribonucleoprotein
hnRNPU	heterogeneous nuclear ribonucleoprotein U
LV	left ventricle
LVEF	left ventricular ejection fraction
miRNA	microRNA
qRT-PCR	quantitative reverse transcription–polymerase chain reaction
RBP	RNA-binding protein
TAVR	transcatheter aortic valve replacement
UTR	untranslated region

Aortic valve stenosis (AVS) constitutes one of the most common degenerative valvular heart diseases, and its prevalence continues to increase with the aging of the population. It is characterized by chronic inflammation with progressive fibrosis and subsequent calcification of the extracellular matrix, which contributes to narrowing of the aortic valve orifice. AVS restricts the blood outflow from the left ventricle (LV) into the aorta and increases LV afterload and workload.[1,2] Depending on the age and operative risk of the patient, severe symptomatic AVS is treated by either surgical aortic valve replacement or transcatheter aortic valve replacement (TAVR). However, predicting whether the accompanying LV dysfunction and heart failure will indeed improve after valve replacement surgery is currently impossible.[3–6] One reason for this is the lack of a mechanistic understanding of why LV function improves in some but not all patients who have undergone valve replacement, although the pressure overload of the LV has been corrected in all patients (assuming adequate prosthesis sizes).

MicroRNAs (miRNAs) are key regulators of gene expression that function by binding to target mRNAs, thereby inducing RNA degradation or blocking translation.[7] Although most miRNAs are found within cells, a few miRNAs are also found extracellularly and are commonly known as circulating or extracellular miRNAs.[8–10] The expression pattern of extracellular miRNAs varies dramatically in different types of biofluids or from different cells. The release of miRNAs is altered under different physiological and pathological conditions; for instance, patients with acute myocardial infarction have significantly higher levels of several plasma miRNAs compared with healthy control subjects.[11,12] However, a reduction in the levels of some circulating miRNAs was also reported in patients with heart failure. As for patients with AVS, previous

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studies indicated that changes in the levels of miRNAs in the valve itself do not necessarily correlate with an alteration of their levels in the plasma.[13] Therefore, circulating miRNAs might affect mainly myocardial remodeling or other systemic dysfunction but not local pathological valvular changes. Some studies have compared the levels of circulating miRNAs in patients with AVS with those in healthy control subjects to identify potential biomarkers and therapeutic targets. Among these circulating miRNAs, some show similar expression patterns at the myocardial level, supporting the idea of a role in the myocardium.[13–15]

Circulating miRNAs are often encapsulated in extracellular vesicles (EVs), which are small membrane-bound vesicles secreted by cells into the extracellular space. EVs can be classified into 3 primary subtypes: exosomes, large EVs (also known as microvesicles), and apoptotic bodies. The cargos within the EVs include lipids, nucleic acids (miRNAs), and proteins.[16–22] It has been shown that EVs are important carriers of bioactive molecules and play essential roles in cell-cell communication between cells. By transferring their cargo to a recipient cell, EVs can modulate the function of the target cells, including the progression of diseases.[20–22]

The expression of miRNAs in patients with AVS with low LV ejection fraction (LVEF) who undergo TAVR has not yet been explored. Thus, delineating the role and function of the differential expression of miRNAs in the pathogenesis and cardiac remodeling in these patients is of great interest. In this study, we aim to identify the circulating miRNAs associated with LVEF after TAVR and to explore the pathological changes in patients with AVS with impaired heart function.

METHODS

Data Availability

The raw proteomic data for this article have recently been published by our group,[21] and we repeated our experiments for this particular study. MiRNA array and apoptosis array data have been deposited in the Gene Expression Omnibus under accession number GSE190576 and in BioProject under accession identifier UB11463488. All further data that support the findings presented here are available on request from the corresponding authors.

Study Approval

A total of 925 patients presenting in our outpatient and emergency departments were enrolled in our study between 2008 and 2014. All clinical samples and measurements were obtained with informed consent from patients after ethics approval by the ethics committee of the University of Bonn (approval 05/12; Supplemental Material).

Human Specimens

Aortic valve specimens were collected from patients undergoing surgical aortic valve replacement for either severe aortic stenosis or aortic regurgitation in cooperation with the Department of Heart Surgery, University Hospital Bonn, Germany. All patients provided written informed consent. The study protocols were approved by the local ethics committee (approval AZ78/17). Total RNA was isolated by a miReasy kit (Qiagen, no. 217004).

Statistical Analysis

Normally distributed continuous variables were presented as the mean±SD. Continuous variables were tested for normal distribution with the use of the Kolmogorov-Smirnov test. Categorical variables are given as frequencies and percentages. For continuous variables, the Student *t* test or Mann-Whitney *U* test was used for the comparison between 2 groups. For the comparison of >2 groups, 1-way ANOVA with Bonferroni correction for multiple-comparisons test was used. The χ^2 test was used for categorical data that resulted from classifying objects. Binary logistic regression was applied to identify factors that were independently associated with miR-122-5p. All tests were 2 sided, and statistical significance was assumed when the null hypothesis could be rejected at $P<0.05$. Statistical analysis was performed with IBM SPSS Statistics version 20 (IBM Inc) and GraphPad Prism 8 (GraphPad Inc).

Methods are given in detail in the Supplemental Material.

RESULTS

Baseline Characteristics and Identification of Circulating MiRNAs in Patients After TAVR

To identify differentially expressed circulating miRNAs in patients with and without LVEF improvement after TAVR, we performed an unbiased quantitative reverse transcription–polymerase chain reaction (qRT-PCR)–based human miRNA array in the screening cohort (Figure 1A). Seven miRNAs (miR-26, miR-885-5p, miR-483-5pa, miR-1274a, miR-192, miR-720, and miR-122-5p; Figure S1B) were shown to be differentially regulated in the group with no LVEF improvement after TAVR. We validated these findings by dividing the validation cohort into 3 groups: patients with robust improvement (>15%), moderate improvement (0%–5%), and no improvement (≤0%) in LVEF at 6 months after TAVR (Table 1). However, our validation experiments found that only miR-122-5p was associated with LVEF improvement and that it is significantly upregulated (fold change, 10; $P=0.005$, false discovery rate adjusted; Figure 1B–1D and Figure S1C) in patients with no LVEF improvement. There were no significant differences with regard to age or sex between the 3 groups. Most patients in this cohort showed increased LVEF (n=71) at 6 months.

Comorbidities that could influence the prognosis of TAVR (eg, pulmonary disease and coronary artery disease) were similar between the groups. Of note, patients with no LVEF improvement had increased levels of troponin, whereas the robust improvement group showed a decrease in troponin levels at 6 months (Figure 1H and Figure S1D). The qRT-PCR results for miRNA

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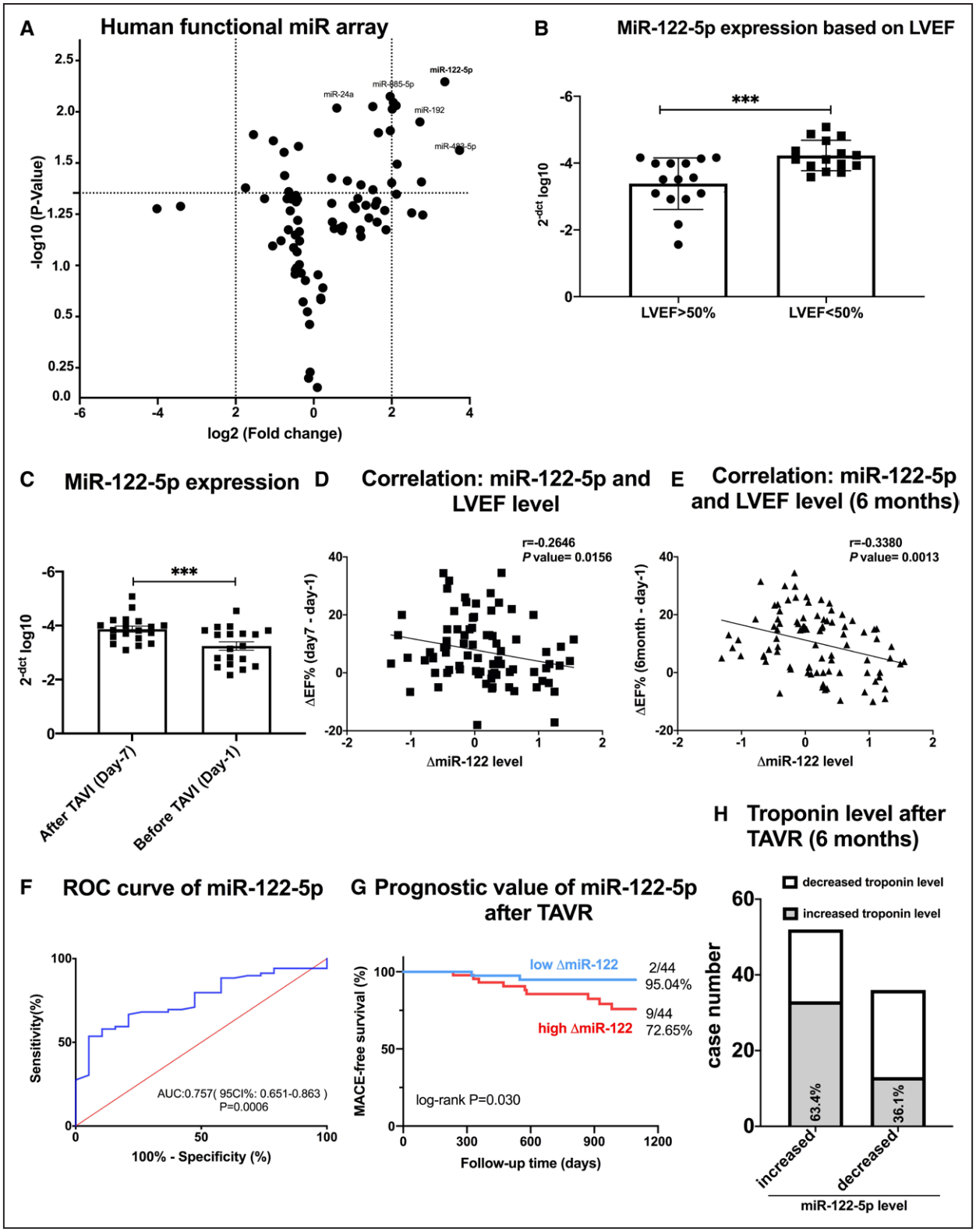

Figure 1. Investigation of circulating miRs in large EVs from patients with and without LV function improvement after TAVR. **A**, Volcano plot showing differentially regulated human microRNAs (miRNAs) in extracellular vesicles (EVs) derived from patient plasma by using an quantitative reverse transcription–polymerase chain reaction (qRT-PCR)–based human miRNA array. Thresholds (gray dotted lines) of a 2-fold change and *P* values (false discovery rate adjusted, –log10 [*P* value]) were used to distinguish the miRNAs of interest. n=3 for day-1, n=3 for day-7 in a group with no left ventricular (LV) ejection fraction (EF) improvement based on echocardiography (no improvement [LVEF ≤0%]). **B**, Expression of circulating EV-associated miR-122-5p was analyzed in 2 groups of patients with LVEF by qRT-PCR. Data represent the mean±SEM (***$P<0.001$; n=15, Student *t* test, 2 tailed, unpaired). miRNA expression and LVEF were measured at (*Continued*)

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Figure 1 Continued. day 7 after transcatheter aortic valve implantation (TAVI). **C**, Expression of circulating EV-associated miR-122-5p was analyzed before and after TAVI (n=19) by qRT-PCR in patients with reduced LVEF. Data represent mean±SEM (***P<0.001, Student t test, 2 tailed, paired). **D**, Correlation analysis of miR-122-5p expression with the change in the level of LVEF from day 1 to day 7 (after TAVI) with a cutoff of LVEF>50% (Spearman coefficient r=–0.2646, P=0.0156 at day 7). **E**, Correlation analysis of miR-122-5p expression with the level of LVEF at the 6-month follow-up (Spearman coefficient r=0.3380, P=0.0013 at 6 months). **F**, Receiver-operating characteristic curve (ROC) analysis for the prediction of LVEF improvement (>0%) at 6 months after transcatheter aortic valve replacement (TAVR) by using the ΔmiR-122-5p level. **G**, Analysis of the prognostic value of miR-122-5p in patients with TAVR. For prognostic value, a Kaplan-Meier survival analysis was applied. Two patients' data were excluded from the follow-up prognostic analysis because of a lack of response. **H**, Change of troponin level in different groups of patients. Comparison of the number of patients with a change in the troponin level when grouped by LVEF improvement after TAVR and by ΔmiR-122-5p level, which represents a change of the miR-122-5p level before and after the TAVR procedure. A Fisher exact test was used for comparison (P=0.0016). Major adverse cardiovascular events (MACEs) included myocardial infarction, heart failure, cardiac arrest, pulmonary embolism, and stroke.

expression in the validation cohort showed that the level of miR-122-5p was significantly higher in the group with no LVEF improvement (Figures 1C and 2). Noticeably, if LVEF became worse in the no-improvement group, the miR-122-5p level also increased after TAVR compared with baseline (Figure 2). Furthermore, we observed that in patients with normal LVEF at baseline, the miR-122-5p level is lower than that of patients with reduced LVEF, which suggests that the miR-122-5p level is associated with LVEF (Figure 2 and Table 1).

Among the miRNAs identified in the arrays, miR-1274a also showed a similar trend, with increased expression in the moderate improvement group (P=0.001), but no significant dysregulation was observed in the no-improvement group (P=0.15; Figure 2 and Figure S1B and S1C). Taken together, our data revealed that miR-122-5p is consistently expressed in patients after TAVR. An intriguing finding was that the increase in the miR-122-5p level negatively correlates with changes in LVEF at day 7 and 6 months after TAVR (r=–0.264, and –0.328, respectively, P=0.0018; Figure 1D and 1E). To assess the diagnostic and prognostic value of miR-122-5p in patients with AVS, a receiver-operating characteristic curve analysis was performed. This analysis suggested that ΔmiR-122-5p (day 7 and day 1 [after TAVR]) was a reliable predictor of LVEF improvement, which reflected the patient's responsiveness to TAVR (area under the curve, 0.757 [95% CI, 0.651–0.863]; P=0.006; Figure 1F). Kaplan-Meier cumulative survival analysis shows that patients with a lower ΔmiR-122-5p displayed a markedly reduced major adverse cardiovascular event–free survival rate (P=0.03) in a 3-year follow-up (Figure 1G). Because we observed an increase in troponin levels, a prominent biomarker for cardiac injury, in the group with no LVEF improvement, we then compared the changes of troponin levels in patients throughout the improvement groups after TAVR. We found that a significantly higher percentage of patients in the improvement group showed reduced troponin levels, whereas in the no-improvement group, the opposite was true (P=0.0038; Figure S1D). A similar trend was observed when the patients were grouped by ΔmiR-122-5p at day 7 (P=0.0016), indicating the positive correlation between the miR-122-5p level and troponin levels (Figure 1H).

Circulating MiR-122-5p Correlates With LVEF in Post-TAVR Patients

Next, we used univariate logistic regression to identify any factors that were associated with LVEF improvement after TAVR. We found the change in stroke volume index (day 7 versus day 1 before TAVR) and transvalvular pressure gradients to be correlated with an improvement of LVEF (odds ratio, 1.038 [95% CI, 1.002–1.075]; P=0.040; odds ratio, 3.149 [95% CI, 1.297–7.645]; P=0.011; Table S1). To further assess all factors related to LVEF improvement, a univariable regression analysis was used (Figure S1C and Table S1). We compared the 2 most differently expressed miRNAs, miR-122-5p and miR-1274a (Figure S1C and Table S1) and found that the stroke volume index change, transvalvular pressure gradient, and troponin change were significantly associated with LVEF improvement when miR-122-5p is differentially regulated. When adjusted for all of these covariates, miR-122-5p was the only factor associated with LVEF improvement (multivariable regression analysis shown in Table 2). On the basis of the aforementioned data, we sought to examine the role of circulating miR-122-5p in patients after TAVR and to investigate its cellular function in regulating cardiovascular pathologies.

MiR-122-5p Is Dysregulated in AVS in Mice and Humans

An increasing number of studies have indicated that the mechanism of AVS and subsequent heart failure depends on cell-to-cell cross-talk.[23–28] To investigate the role of miR-122-5p in cardiac function, we used a graded wire injury model of AVS in mice, as previously described[27] (Figure 3A). The AVS phenotype was confirmed by measuring the longitudinal peak velocity on day 1 and week 6 after the injury (Figure 3B). qRT-PCR analysis revealed that miR-122-5p is highly upregulated in mice with stenosis (with reduced LVEF; Figure 3C). However, the LVEF and LV mass were inversely regulated in these mice, suggesting that there is an AVS-induced decrease of LVEF (Figure 3D and 3E). Valvular thickening was analyzed with hematoxylin and eosin staining, which suggested that wire injury increases stenosis, agreeing with

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Table 1. Baseline Characteristics of the Study Population

Variable	Total	No improvement	Moderate improvement	Robust improvement	*P* value
Total population, n	90	19	37	34	…
Age, y	80.0±6.2	81.7±76.3	79.1±6.2	79.9±6.1	0.35
Male sex, %	68.9	12 (63.2%)	78.4	61.8	0.27
NYHA level	3.2±0.5	3.3±0.5	3.1±0.5	3.2±0.5	0.45
Body mass index, kg/m^2	25.1±5.4	25.2±3.7	24.3±6.7	25.8±4.9	0.56
Diabetes mellitus, n (%)	21 (23.3)	7 (36.8)	7 (18.9)	7 (20.6)	0.30
Coronary artery disease, n (%)	59 (65.66)	13 (68.4)	24 (64.9)	22 (64.7)	0.96
COPD, n (%)	25 (27.8)	6 (31.6)	8 (21.6)	11 (32.4)	0.56
Atrial fibrillation, n (%)	47 (52.2)	12 (53.2)	19 (51.4)	16 (47.1)	0.54
Previous myocardial infarction, n (%)	14 (15.6)	5 (16.3)	5 (13.5)	4 (11.8)	0.35
EuroSCORE II	10.5±7.7	9.8±7.4	10.1±8.7	12.0±8.4	0.48
STS score	8.0±5.9	8.3±5.8	6.7±4.7	9.0±6.9	0.11
eGFR, mL/min	53.6±18.3	54.2±15.8	51.6±20.0	53.0±15.6	0.96
Mild-severe mitral regurgitation, n (%)	51 (56.7)	11 (57.9)	24 (64.9)	16 (47.1)	0.32
Laboratory parameters					
CK-MB before TAVR, %	2.2±1.6	2.6±2.5	2.2±1.3	2.1±1.1	0.56
CK-MB after 7 d, %	1.7±1.1	1.7±1.3	1.7±1.0	1.7±1.2	0.97
CK-MB after 6 mo, %	2.2±1.7	2.3±1.7	2.1±1.7	2.2±1.9	0.90
CK-MB change (after 6 mo−before), %	−0.03±1.38	−0.16±0.73	0.13±1.31	−0.14±1.70	0.66
Troponin before TAVR, ng/mL	0.07±0.10	0.03±0.02	0.08±0.13	0.08±0.10	0.22
Troponin after day 7, ng/mL	0.16±0.33	0.25±0.56	0.15±0.30	0.13±0.13	0.44
Troponin after 6 mo, ng/mL	0.05±0.12	0.09±0.09	0.06±0.06	0.02±0.02	0.11
Troponin change (after 6 mo−before), ng/mL	0.02±0.14	0.06±0.16	−0.02±0.14	−0.05±0.10	0.01
Baseline echocardiography					
LVEF before TAVR, %	35.3±7.5	38.5±4.1	35.5±6.6	33.4±9.3	0.18
LVEF after 7 d, %	42.8±12.1	35.6±7.0	40.9±9.7	49. 0±14.0	0.00
LVEF post 6 mo, %	45.7±11.0	34.6±5.1	43.6±7.4	54.3±10.2	0.00
LVEF change (after 6 mo−before), %	10.4±10.4	-3.9±3.6	8.1±4.6	20.9±5.2	0.00
LVEDVi before TAVR, mL	80.1±31.3	81.7±30.1	82.3±34.4	76.9±29.9	0.79
LVEDVi after 7 d, mL	77.1±31.4	87.5±37.0	77.3±34.8	71.3±24.1	0.24
LVEDVi after 6 mo, mL	65.6±27.2	78.9±35.4	64.8±26.1	58.2±19.1	0.03
LVEDVi change (after 6 mo−before), mL	12.70±25.8	-2.8±23.7	-18.4±26.1	-13.7±26.0	0.13
SVi before TAVR, mL/m^2	29.1±11.6	32.0±11.1	28.7±10.7	27.8±12.7	0.46
SVi after 7 d, mL/m^2	32.2±14.5	32.7±17.8	28.5±11.9	34.7±14.3	0.31
SVi after 6 mo, mL/m^2	28.4±9.4	25.7±9.0	26.8±10.7	31.4±7.5	0.31
SVi change (after 6 mo−before), mL/m^2	−0.8±11.6	−5.7±−2.3	−2.3±9.7	3.5±12.2	0.02
IVSd before TAVR, mm	1.19±0.23	1.21±0.23	1.23±0.24	1.1±0.22	0.24
IVSd after 6 mo, mm	1.16±0.23	1.12±0.19	1.18±0.19	1.16±0.29	0.68
PWd before TAVR, mm	1.16±0.24	1.16±0.21	1.15±0.22	1.17±0.29	0.93
PWd after 6 mo, mm	1.19±0.24	1.1±0.19	1.21±0.22	1.22±0.29	0.39
Transvalvular gradient >40 mm Hg, n (%)	30 (38.0)	4 (21.1)	9 (31.0)	17 (54.8)	0.04

Baseline demographic, laboratory, and echocardiographic parameters of the validation study population are shown. *P* values reflect the comparison among 3 different groups.

CK-MB indicates creatine kinase-MB; COPD, chronic obstructive pulmonary disease; eGFR, estimated glomerular filtration rate; EuroSCORE II, European System for Cardiac Operative Risk Evaluation II; IVSd, interventricular septal diameter; LVEDVi, left ventricular end-diastolic volume index; LVEF, left ventricular ejection fraction; NYHA, New York Heart Association; PWd, posterior wall diameter; STS, Society of Thoracic Surgeons; SVi, stroke volume index; and TAVR, transcatheter aortic valve replacement.

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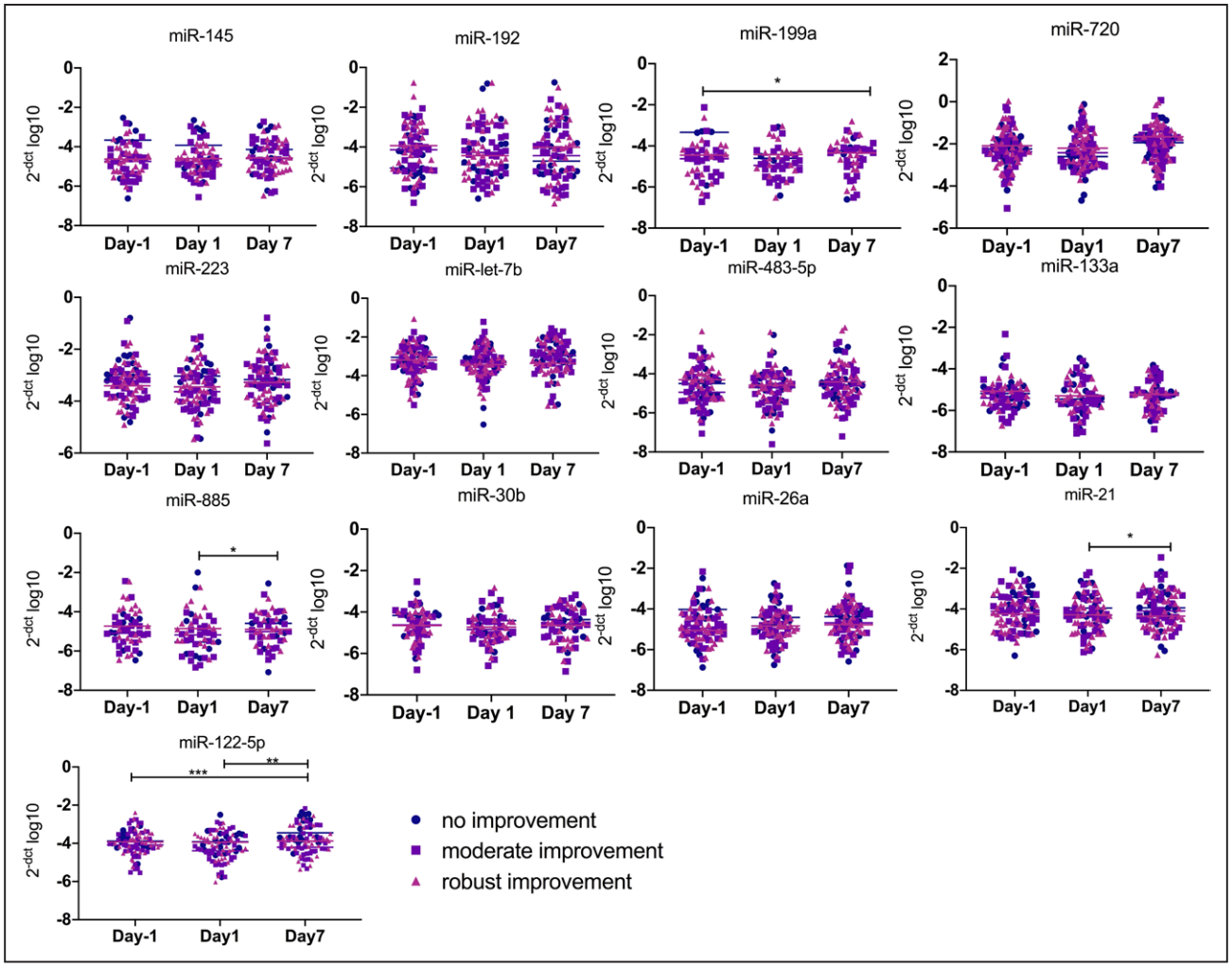

Figure 2. Analysis of miR expression in patients with low LVEF after transcatheter aortic valve replacement. MicroRNA (miR) expression was analyzed in circulating extracellular vesicles (EVs) that were isolated from 90 patients with transcatheter aortic valve replacement (TAVR) with (echocardiographically determined) no improvement (n=19), moderate improvement (n=37), and robust improvement (n=34) of left ventricular ejection fraction (LVEF) at 3 different time points (day 1 [before TAVR], day 1 [after TAVR], and day 7 [after TAVR]). Values were normalized to cel-miR-39 and were expressed as $2^{-[CT(miR)-CT(cel\text{-}miR\text{-}39)]}$ $\log_{10}$. Data are presented as mean±SEM (*P<0.05, **P<0.01, ***P<0.001, by 2-way ANOVA with Bonferroni correction for multiple-comparisons test).

the echo data (peak velocity; Figure S2B). To rule out that wire injury induces apoptosis in the myocardium, we used a terminal deoxynucleotidyl transferase dUTP nick-end labeling assay in mice that had undergone an AVS operation. Compared with the positive control (treated with DNase 100 U), no terminal deoxynucleotidyl transferase dUTP nick-end labeling–positive cells were observed, indicating that AVS mice do not show any induction of apoptosis in the myocardium (Figure S2C). Together, our data revealed that LVEF was inversely regulated with miR-122-5p in a murine model of AVS.

Table 2. Multivariable Logistic Regression for Factors Associated With LVEF Improvement

	Significance	Exp(B) odds ratio	95% CI
Gradient, mm Hg	0.123	3.769	0.698–20.352
MiR-122-5p change	0.019*	0.134	0.025–0.720
MiR-1274a change	0.726	1.163	0.499–2.710
SVi change, mL/m^2	0.901	1.004	0.948–1.063
Troponin change, ng/mL	0.190	0.110	0.004–2.991
Previous myocardial infarction	0.972	0.968	0.150–6.240

Change defined as the parameter at day 7 minus day 1.

Exp (B) indicates exponentiation of the B coefficient; LVEF, left ventricular ejection fraction; miR, microRNA; and SVi, stroke volume index.

*P<0.05.

An increase of fractional shortening in stenotic mice after week 6 revealed a time-dependent decrease of contractility of the cardiomyocytes and impairment of cardiac output on AVS (Figure 3F and 3G). Because the level of miR-122-5p was found to correlate with LV function, we examined whether the LV dysfunction was caused by volume or pressure overload, which could be attributable to cardiac hypertrophy. At 6 weeks after AVS induction, only a modest increase in anterior and posterior wall thicknesses was observed. Because of

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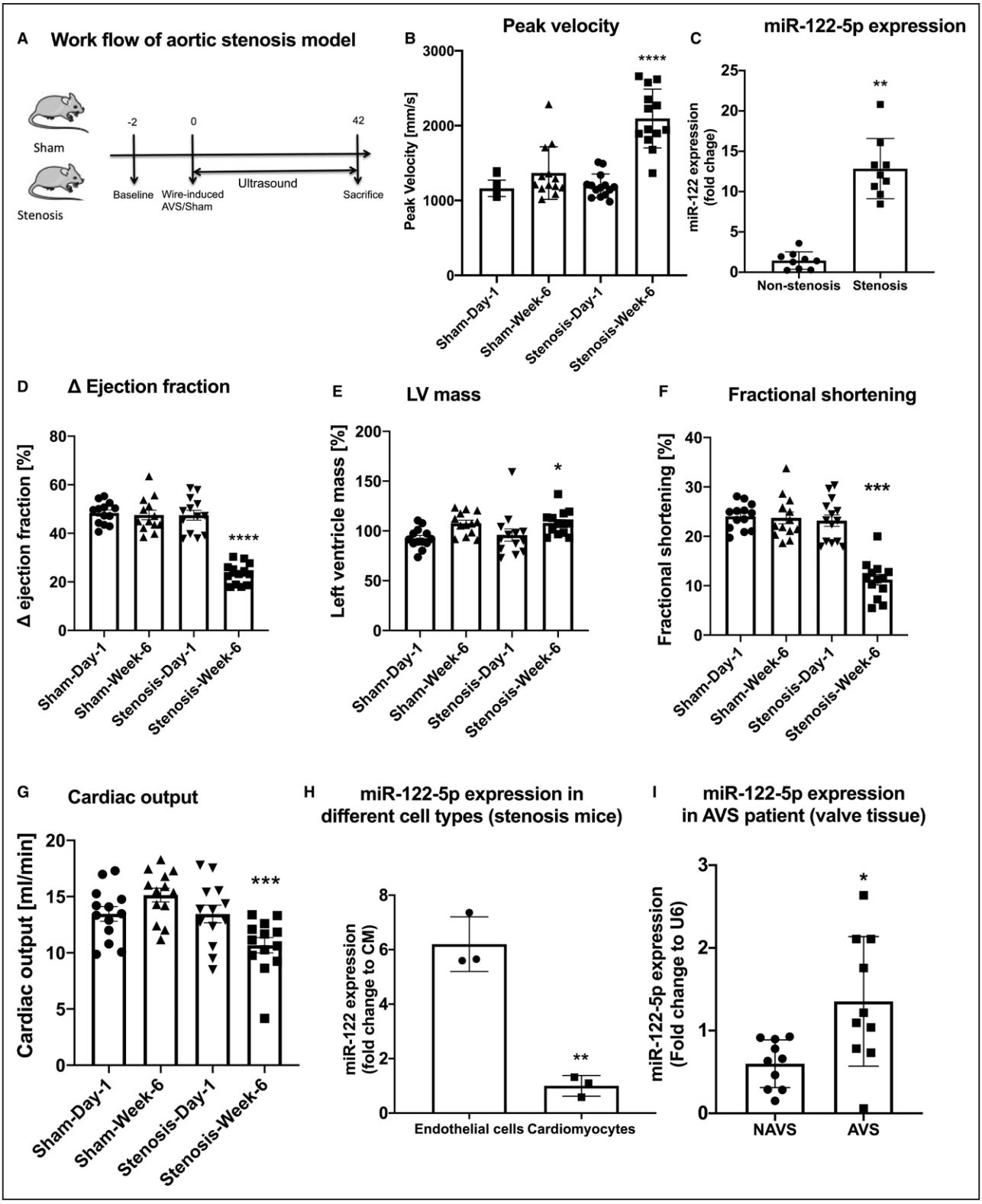

Figure 3. MiR-122-5p is differentially regulated in a murine model of AVS and patients with AVS.
A, Timeline of the operation used to produce aortic valve stenosis (AVS) in a murine model. Induction of AVS in mice is achieved by inserting a coronary wire into the left ventricle (LV) and rotating it. **B**, Peak velocities over the aortic valve after a sham operation or operation creating severe stenosis at day 1 and week 6 after the surgery. Statistical significance was shown between sham day 1 and sham week 6 and stenosis day 1 and stenosis week 6 (****$P<0.0001$; sham n=13; stenosis n=13; SEM; 1-way ANOVA with Bonferroni multiple-comparisons test). **C**, MicroRNA (MiR)-122-5p expression in stenotic murine hearts. Data were normalized to *RNU6B* (U6) and depicted as ±SEM (**$P<0.01$; n=9; Student *t* test, unpaired, 2 tailed). **D**, Change in ejection fraction after sham operation or operation creating severe stenosis at day 1 and week 6 compared with before the surgery. Statistical significance was shown between sham day 1 and sham week 6 and stenosis day 1 and (*Continued*)

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Figure 3 Continued. stenosis week-6 (****P<0.0001; sham n=13; stenosis n=13; SEM; 1-way ANOVA with Bonferroni multiple-comparisons test). **E**, LV mass after a sham operation or operation creating severe stenosis at day 1 and week 6. Statistical significance was shown between sham day 1 and sham week 6 and stenosis day 1 and stenosis week 6 (*P<0.5; sham n=13; stenosis n=13; SEM; 1-way ANOVA with Bonferroni multiple-comparisons test). **F**, Fractional shortening over the aortic valve after a sham operation or operation creating severe stenosis at day 1 and week 6 (***P<0.001; sham n=13; stenosis n=14; SEM; 1-way ANOVA with Bonferroni multiple-comparisons test). **G**, Cardiac output over the aortic valve after a sham operation or operation creating severe stenosis at day 1 and week 6 (***P<0.001; sham n=13; stenosis n=13; SEM; 1-way ANOVA with Bonferroni multiple-comparisons test). **H**, MiR-122-5p expression, as determined by quantitative reverse transcription–polymerase chain reaction (qRT-PCR), in different cell types isolated from stenotic murine hearts (**P<0.01; stenosis n=3; SEM; Student t test, unpaired, 2 tailed). **I**, miR-122-5p expression, as determined by qRT-PCR, in valvular tissue of patients with AVS and without stenosis (NAVS; *P<0.05; n=10; with stenosis n=10; SEM; Student t test, unpaired 2 tailed). CM indicates cardiomyocytes.

the thickening of both walls, we interpret this to be concentric hypertrophy, which is caused by pressure overload (not volume overload, which would cause eccentric hypertrophy; Figure S2D and S2E). To translate these findings to cardiac cells, we isolated cardiac endothelial cells (ECs) and cardiomyocytes from stenotic mice to quantify the cell-specific expression of miR-122-5p at 6 weeks after AVS injury. We found that cardiomyocytes express a relatively lower level of miR-122-5p compared with ECs (Figure 3H). Moreover, miR-122-5p expression was upregulated in the patient-derived valvular tissue samples explanted from patients with AVS (Figure 3I). These data indicate that AVS is associated with increased valvular miR-122-5p expression and may play a role in cardiac function both in mice and humans.

MiR-122-5p Is Encapsulated in EVs

We aimed to determine the plasma component (exosomes, large EVs, or vesicle-free plasma) in which miR-122-5p is primarily found. To achieve this, large EVs, exosomes, and vesicle-free plasma were collected by use of a series of centrifugation/ultracentrifugation steps, as previously published.[12,20–22,29] MiR-122-5p levels were analyzed in these subcompartments of patient blood samples after TAVR (Figure 4A). To explore whether circulating extracellular miR-122-5p is encapsulated in plasma EVs or bound to protein complexes, we performed a vesicle-RNA degradation assay (Figure 4B). We found that the digestion of proteins (using proteinase K), before treatment with RNase, did not affect miR-122-5p levels. In contrast, treatment with Triton X-100, which acts as a detergent to disrupt the phospholipid membrane of vesicles, before treatment with RNase, led to near-complete degradation of miR-122-5p. These findings indicate that circulating extracellular miR-122-5p is incorporated primarily into EVs, which protect the miRNAs from resident nucleases (Figure 4B). The characterization of plasma- and cell-derived EVs was performed according to the current guidelines of the International Society of Extracellular Vesicles through nanoparticle tracking analysis, transmission electron microscopy, and immunoblotting for vesicular markers, including tetraspanins (Figure 4C–4E and Figure S3A–S3C). Together, these data suggest that miR-122-5p is shuttled into target cells through large EVs.

Vesicular MiR-122-5p Regulates Apoptosis of Recipient Cardiomyocytes

Next, we explored whether ECs can transfer miR-122-5p into cardiomyocytes and regulate its function. We speculated that the dysfunction of cardiomyocytes is crucial to the decrease of LV function in the murine AVS model (decreased fractional shortening and lower cardiac output correlated to LV mass). A comparative study using copy-number analysis revealed that cardiomyocytes and cardiomyocyte-derived EVs expressed miR-122-5p less abundantly than ECs and their corresponding EVs (Figure 5A). We also aimed to determine the function of EV-incorporated miR-122-5p in target cardiomyocytes by knocking down miR-122-5p with an inhibitor (Figure 5B). To dissect the role of EV-incorporated miR-122-5p in intercellular communication, we established a coculture model. We transfected ECs with miR-122^{control}, miR-122^{inhibitor}, or miR-122^{mimic} and cocultured them with cardiomyocytes under different conditions (Figure 5C). Most important, the coculture experiments revealed that miR-122-5p is differentially expressed in EC-cardiomyocytes coculture compared with cardiomyocyte control alone, implying that there is transfer of miR-122-5p from ECs to cardiomyocytes. This differential export of miR-122-5p is higher in EC$^{miR\text{-}122\text{-}mimic}$-treated cardiomyocytes than EC^{control}- or EC^{Inhibitor}-treated cardiomyocytes, suggesting that EV-mediated cell-to-cell transfer of miR-122-5p could play a significant role in cellular function (Figure 5C).

To analyze whether intercellular transfer of miR-122-5p directly regulates the function of target cardiomyocytes, cell viability and apoptosis were quantified through the use of an MTT assay and flow cytometry after Annexin-V-7AAD staining, respectively (Figure 5D and 5E). These data demonstrated that EV-mediated transfer of miR-122-5p regulates the viability and apoptosis of target cardiomyocytes by exerting proapoptotic effects.

Vesicular MiR-122-5p Interacts With Heterogeneous Nuclear Ribonucleoprotein U

Recent studies have reported a short but specific sequence motif in miRNAs that regulates their packaging into EVs.[30,31] To examine whether the packaging of miR-122-5p into EVs is mediated by RNA-binding proteins

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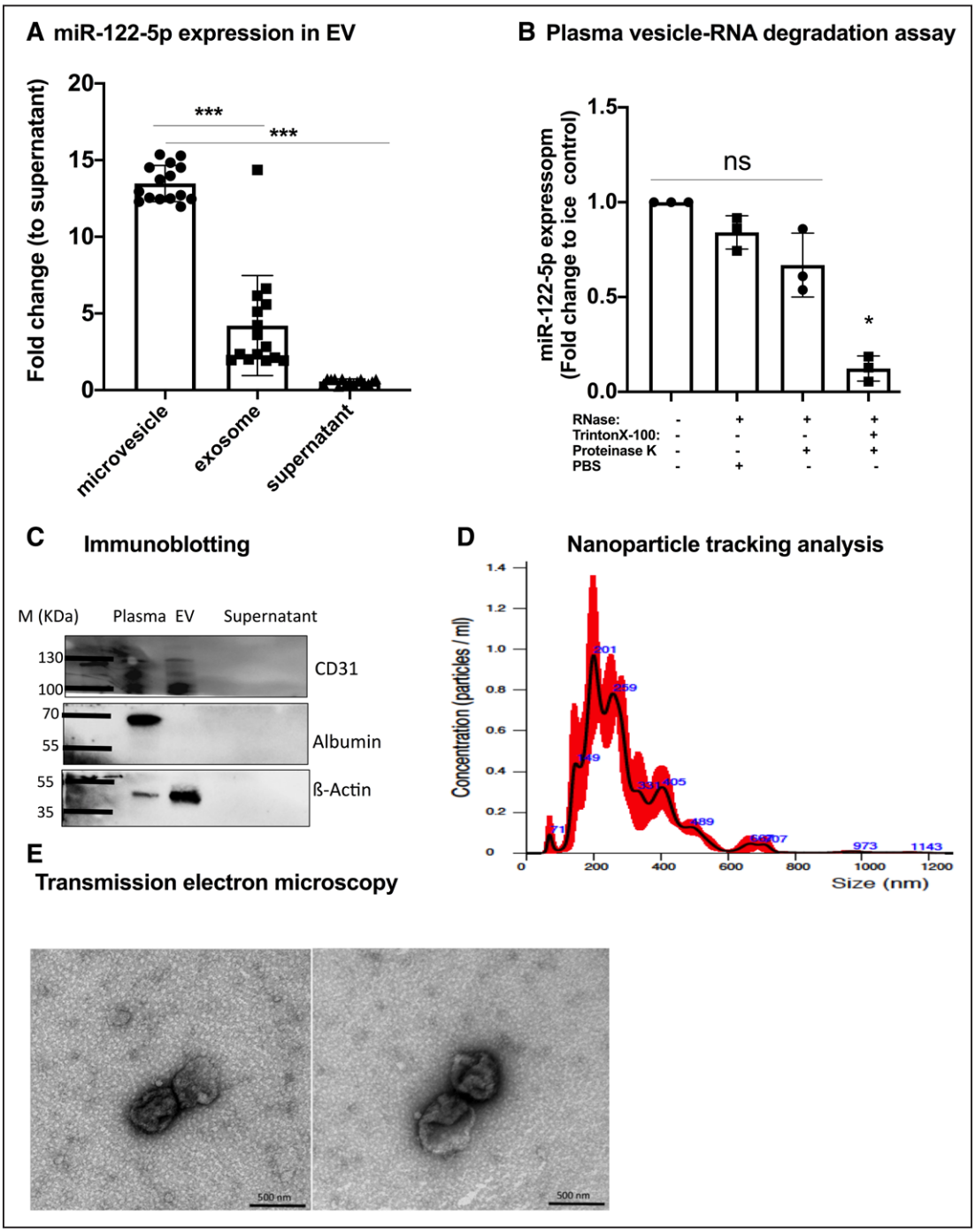

Figure 4. MiR-122-5p is encapsulated in the large extracellular vesicles.
A, MicroRNA (miR)-122-5p expression was assessed in different extracellular vesicle (EV) populations isolated from patient plasma by quantitative reverse transcription–polymerase chain reaction (***P<0.001; n=15, by 1-way ANOVA with Bonferroni correction for multiple-comparisons test). Cel-miR-39 was used for normalization. EVs and exosomes were isolated by centrifugation at 20 000g and 100 000g, according to the protocol published by our group and others.[20,21,29–31] **B**, Vesicle-RNA degradation assays. Plasma-derived EVs were treated in parallel with different conditions followed by RNase A digestion. MiR-122-5p was quantified by quantitative reverse transcription–polymerase chain reaction (*P<0.05 compared with the untreated group; not significant [ns] n=3; 1-way ANOVA with Bonferroni correction for multiple-comparisons test). **C**, Western blot analysis of the expression of 3 EV and cell markers. **D**, Nanoparticle tracking analysis was used to determine the diameter (≈200–700 nm) and concentration of EVs in the cell culture supernatant. **E**, Transmission electron microscopy image (85 000x magnification) of pelleted large EVs (diameter, ≈300–700 nm) derived from the plasma of patients with aortic valve stenosis (AVS)/without AVS (NAVS). Large EVs derived from the plasma of patients with NAVS (**left**) and AVS (**right**).

(RBPs), we used publicly available online tools to identify protein-interaction partners for miR-122-5p (Figure 6A).[32–34] Sixty-four candidates were overlapping between the 2 platforms, and the top 10 protein candidates were all annotated as RBPs. The multifunctional RBP hnRNPU (heterogeneous nuclear ribonucleoprotein U)

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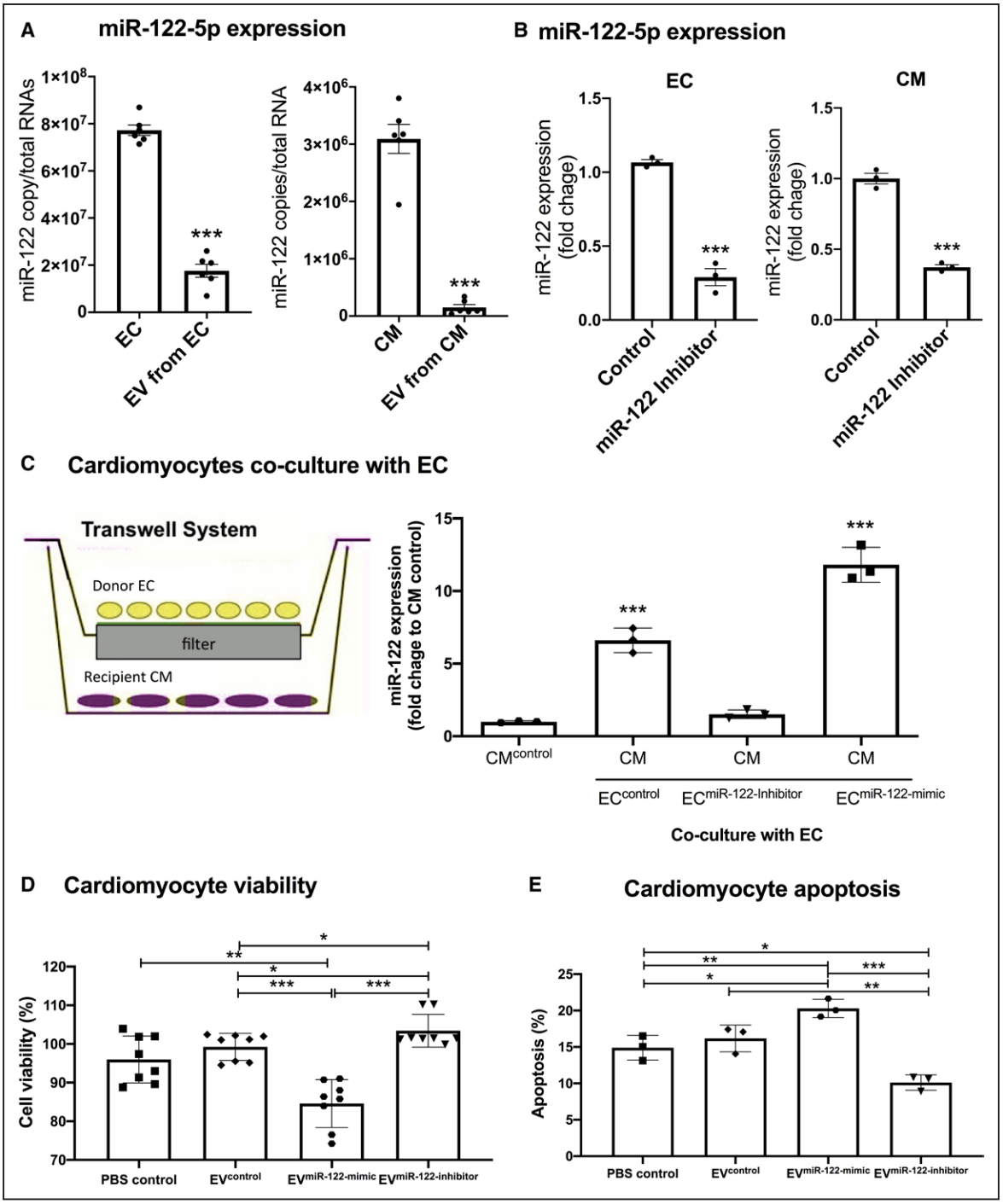

Figure 5. MiR-122-5p regulates apoptosis of recipient cardiomyocytes through vesicular transfer.
A, Expression of microRNA (miR)-122-5p was measured in endothelial cells (ECs) and cardiomyocytes (CMs) with their corresponding extracellular vesicles (EVs). A copy-number assay was performed to determine miR-122-5p copies per total RNA. Data are depicted as Student *t* test. (***$P<0.001$; n=6, 2 tailed, unpaired). Expression of miR-122-5p was normalized to the internal control gene *RNU6*. **B**, Endothelial cells and CMs were transfected with miR-122-5p inhibitor and siRNA control. Knockdown efficiency of miR-122-5p was quantified by use of a miR-inhibitor in ECs and CMs. Expression of miR-122-5p was normalized to *RNU6B*. Data are depicted as Student *t* test (***$P<0.001$; n=3, 2 tailed, unpaired). **C**, EC-CM coculture experiments. ECs were transfected with control, miR-122-5p inhibitor, or miR-122-5p mimic and coincubated with CMs under similar treatment conditions to quantify the differential miR-122-5p transfer through EVs. Cycle of threshold values were normalized to *RNU6B* and expressed as fold change with 1-way ANOVA with Bonferroni multiple-comparisons test (*$P<0.001$; n=3). **D** and **E**, Cell viability and apoptosis measurement. CMs were incubated with EVs derived from ECs transfected with siRNA control, miR-122-5p mimic, or miR-122-5p inhibitor or a PBS control. An apoptosis assay was performed through Annexin-V/7AAD flow cytometry. Annexin-V–positive cells were defined as indicating apoptosis. Data are represented as the percentage of apoptotic cells and analysis by 1-way ANOVA with Bonferroni multiple-comparisons tests (*$P<0.05$, **$P<0.01$, ***$P<0.001$; n=3).

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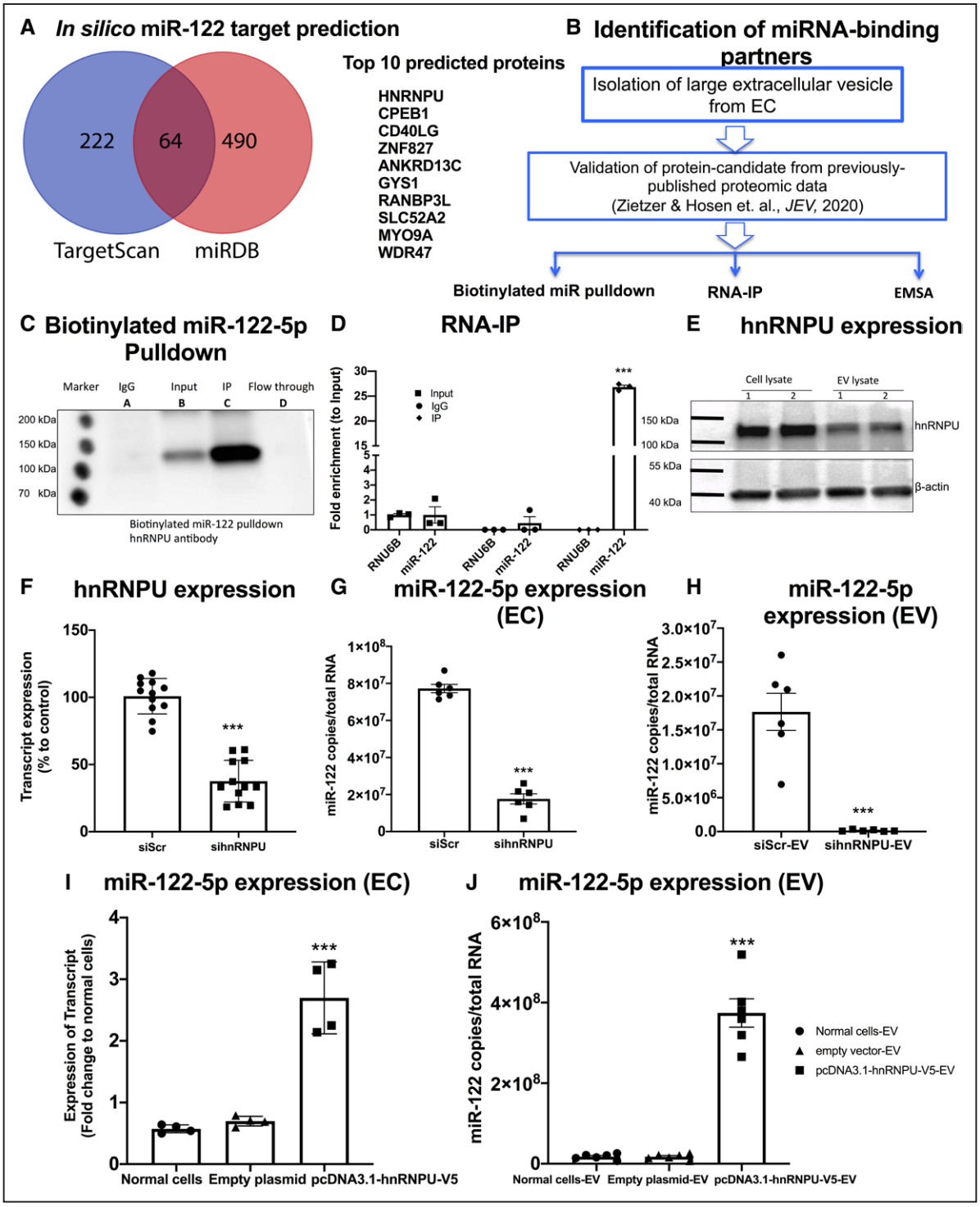

Figure 6. Interaction between miR-122-5p and hnRNPU mediates miR packaging into EVs.
A, Venn diagram of the in silico prediction of microRNA (miR)-protein interaction partners by using publicly available tools (TargetScan and miRDB); the top 10 protein-interacting candidates are shown. **B**, Workflow for identifying protein candidates in extracellular vesicles (EVs) from our previously published mass spectrometry data.[21] The identified protein candidates from mass spectrometric analysis were validated by RNA pulldown, RNA immunoprecipitation (IP), and electrophoretic mobility shift assay (EMSA). **C**, Biotinylated miRNA pulldown followed by Western blotting, with the enriched RNA-binding proteins (RBPs) identified via mass spectrometry using an antibody against hnRNPU (heterogeneous nuclear ribonucleoprotein U) and a biotin-labeled probe of miR-122-5p. The respective immunoglobulin G (IgG) antibody from the same species was used as an IP negative control. Enrichment was confirmed with input (10%) from the cellular lysates. **D**, Cross-linked RNA-IP experiments followed by quantitative reverse transcription–polymerase chain reaction (qRT-PCR) quantification to confirm the reciprocal binding of hnRNPU to miR-122-5p in human endothelial cells. From RNA-IP experiments, strong binding of hnRNPU was observed compared with control IgG (negative control). Quantification was performed by normalizing the values with the inputs (10%). (*Continued*)

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Figure 6 Continued. Data represent mean±SEM (***$P<0.001$; n=3; Student *t* test). **E**, Western blotting experiments confirmed the enrichment of hnRNPU in endothelial cells (ECs) and their EVs. **F**, Silencing of hnRNPU in ECs by use of siRNAs against hnRNPU compared with a scrambled siRNA control. HnRNPU expression was analyzed in ECs by qRT-PCR (***$P<0.001$;n=12; Student *t* test). **G** and **H**, MiR-122-5p expression in ECs and the corresponding EVs on hnRNPU silencing via siRNAs. MiR-122-5p expression was assessed in target ECs that were treated with PBS and EVs using copy-number analysis (***$P<0.001$; n=6; 1-way ANOVA with Bonferroni multiple-comparisons test). **I** and **J**, Plasmid-mediated stable overexpression of hnRNPU in ECs and the subsequent quantification of miR-122-5p in ECs and the corresponding EVs by using copy-number analysis (***$P<0.001$; n=6; 1-way ANOVA with Bonferroni multiple-comparisons test).

was the top candidate identified in silico (Figure 6A and Tables S2 and S3).

To elucidate the mechanism of the packaging of miR-122-5p into EVs, we previously performed mass spectrometric analysis of EVs to detect EV-enriched proteins, and we detected >3000 proteins expressed within EVs generated from ECs.[21] A significant portion of the protein candidates that were identified were annotated as RBPs, according to SwissProt.[21] In particular, heterogeneous nuclear ribonucleoproteins (hnRNPUs) were expressed prominently in isolated large EVs.[21,35,36] To validate these findings and to determine the relationship between miR-122-5p and RBPs, we performed multiple experiments to detect the miR-122-5p–protein interaction (Figure 6B). RNA pull-down followed by immunoblotting identified hnRNPU as a binding partner of miR-122-5p (Figure 6C). RNA immunoprecipitation followed by qRT-PCR showed that miR-122-5p binds strongly to hnRNPU (Figure 6D) compared with *RNU6B*, which suggests that hnRNPU is a direct binding partner of miR-122-5p.

To further confirm the interaction, we performed an electrophoretic mobility shift assay and demonstrated that biotin-labeled miR-122-5p directly binds to recombinant hnRNPU, whereas binding was not observed between unlabeled miR-122-5p and recombinant hnRNPU or biotinylated miR-122-5p alone, serving as a negative control (Figure S4A). These data indicate that hnRNPU is a binding partner of miR-122-5p and that it may facilitate its incorporation into large EVs.

HnRNPU May Control the Loading of MiR-122-5p Into EVs

hnRNPs are a large family of RBPs that mediate the loading of noncoding RNAs into EVs.[21,30,31] Multiple studies revealed that hnRNPU-mediated shuttling of miRNAs into small EVs is controlled by a specific sequence motif in the miRNAs.[21,31,32] A search of sequence motifs with the motif enrichment tool CentriMo with ZOOPs[37] revealed that a 4- to 8-nucleotide sequence (*AAMRUGCU*) known to bind hnRNPU with miRNAs is present in miR-122-5p.[21] This sequence has been identified as the key component for hnRNPU/hnRNP binding and packaging of miRNAs into EVs.[21,31–36] Our experiments revealed that hnRNPU is expressed predominantly in ECs and their corresponding EVs (Figure 6E). To address whether hnRNPU directly affects the expression of miR-122-5p, we quantified miR-122-5p by using copy-number analysis in ECs and the corresponding EC-derived vesicles after the loss of hnRNPU (Figure 6F). The copy-number analysis revealed that silencing hnRNPU impaired the expression of miR-122-5p in donor ECs and that its packaging is significantly downregulated in EVs on silencing of hnRNPU (Figure 6G and 6H). An intriguing finding was that hnRNPU silencing reduced cellular expression of miR-122-5p, which was revealed using immunofluorescence with a Cy3-labeled miR-122-5p and an antibody against hnRNPU compared with the siRNA control (Figure S4B).

To confirm the role of hnRNPU-mediated export of miR-122-5p, hnRNPU was overexpressed in ECs by performing a transfection of the mammalian expression vector pcDNA 3.1-hnRNPU-V5. Overexpression of hnRNPU led to upregulation of miR-122-5p, which was quantified by qRT-PCR (relative quantification and copy-number analysis) compared with cells without plasmid transfection or empty plasmid transfection in donor ECs and their corresponding EVs (Figure 6I and 6J). These data indicate that hnRNPU contributes to the sorting and packaging of miR-122-5p into large EVs.

MiR-122-5p Regulates Apoptosis of Cardiomyocytes in a BCL2-Dependent Manner

To obtain further insights into the function of miR-122-5p, we performed an unbiased human apoptosis array to quantify the genes involved in the regulation of apoptosis on silencing of miR-122-5p in EV donor cells and subsequent treatment with the corresponding EVs to examine apoptotic gene expression in recipient cardiomyocytes. A number of genes that were differentially regulated are known as critical regulators of apoptosis in cardiomyocytes. Among them, the antiapoptotic genes *BCL2*, *NOD2*, and *BIRC8* were upregulated on miR-122 knockdown but not in a statistically significant manner (Figure 7A and 7B and Figure S5A). Independent qRT-PCR analysis revealed that these genes were indeed differentially regulated in cardiomyocytes and in their corresponding EVs (Figure S5A).

Given that BCL2 is a central player in apoptosis, we sought to further characterize its regulation by miR-122-5p. A recent study revealed that several miRNAs have the potential to control the translation or degradation of *BCL2* mRNA by binding to the 3′ untranslated region (UTR). To this end, we used computational approaches to identify miRNAs that could negatively regulate *BCL2* expression by binding to its 3′ UTR. As

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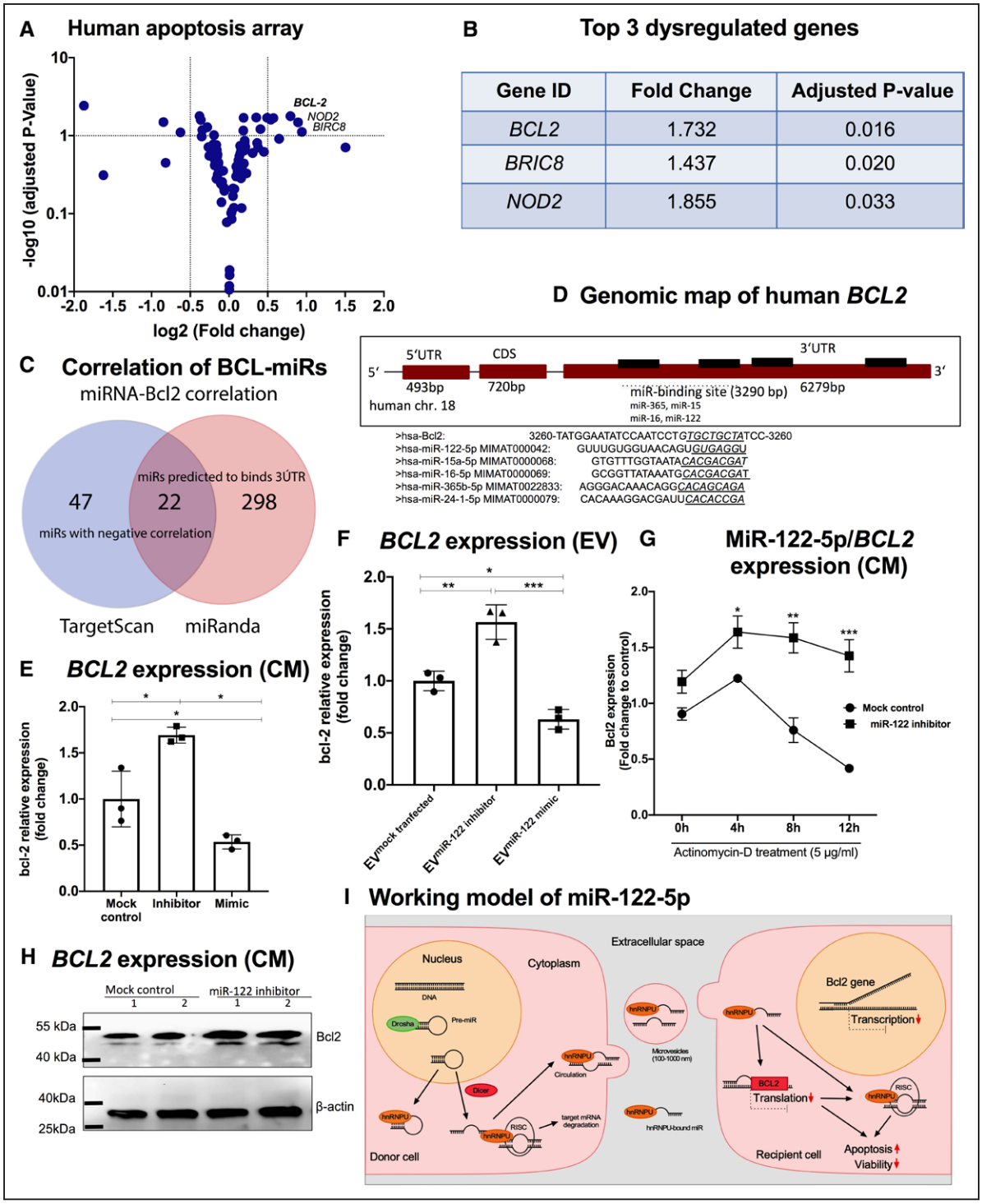

Figure 7. MiR-122-5p regulates apoptosis by inducing *BCL2* and other apoptotic genes.
A, A Taqman-based human apoptosis array with known apoptosis-regulated genes comparing expression of mRNAs in control- vs miR-122-5p inhibitor–transfected human cardiomyocytes (CMs) was used to assess the levels of apoptosis-related genes present in the cells (n=3; SEM; false discovery rate [FDR] adjusted). **B**, The top 3 upregulated genes as detected by Taqman-based human apoptosis array (n=3; SEM; FDR adjusted). **C**, Venn diagram showing a prediction of miR-*BCL2* interactions based on the correlation coefficient values between *BCL2* mRNA and miRNAs in publicly available miRNA–protein target prediction databases. Expression data of 319 miRNAs were available for the 48 cell lines; expression of 69 miRNAs was shown to be negatively regulated with *BCL2*, and 320 target miRNAs were predicted to directly bind to *BCL2* mRNA at the 3′ untranslated region (UTR). MiR-122-5p belongs to 22 common miRs that are predicted to be an interaction partner of *BCL2* by both of the publicly available online prediction tools with high correlation coefficients. **D**, Schematic representation of the *BCL2* gene with predicted binding sites for hsa-miR-122-5p, hsa-miR-15a-5p, hsa-miR-16a-5p, hsa-miR-365b-5p, and hsa-miR-24 in the 3' UTR. **E** and **F**, Expression data for *BCL2* mRNA in CMs and the corresponding extracellular vesicles (EVs) on transfection with miR-122-5p inhibitor or mimic compared with a control (*$P<0.05$, **$P<0.01$, ***$P<0.001$; n=3; 1-way ANOVA with Bonferroni multiple-comparisons test). (*Continued*)

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Figure 7 Continued. G, Expression of *BCL2* mRNA in CMs on actinomycin D treatment (to block transcription of mRNAs) at different time points on transfection with miR-122-5p inhibitor or control RNA to assess the stability of the transcript (**P*<0.05, ***P*<0.01, ****P*<0.001; n=3; 1-way ANOVA with Bonferroni multiple-comparisons test). **H**, Western blotting for BCL2 protein after transfection with miR-122-5p inhibitor and control (n=3). **I**, Model proposed for miR-122-5p action in which it is shuttled to the recipient cells via hnRNPU (heterogeneous nuclear ribonucleoprotein U)–miR-122-5p conjugates to regulate apoptotic signaling networks such as BCL2 to control the apoptosis of CMs. ID indicates identification.

presented in Figure 7C and Figure S5B, the Venn diagram and table show that different miRNAs are correlated on the basis of the correlation coefficient values between miRNA expression and *BCL2* mRNA expression across 48 cell lines.[31,32] Of those analyzed, 319 miRNAs were shown to bind to the 3′ UTR of *BCL2* mRNA, and 69 miRNAs demonstrated a significant negative correlation with *BCL2* expression in TargetScan and miRanda.[31,32] MiR-122-5p demonstrated a negative correlation as a potential miRNA for targeting BCL2 (Figure 7C, Figure S5B, and Table S4). However, when the binding sites for miRNAs in the *BCL2* gene were analyzed, we found that miR-122-5p has a complementary binding site in the 3′ UTR, as do several other miRNAs (miR-365b, miR-15/16, and miR-24-5p; Figure 7D), suggesting that miR-122-5p interacts with *BCL2*. Thus, we attempted to identify the biological effects of the repression of *BCL2* by miR-122-5p that regulate the cellular function of cardiomyocytes. To examine the effects of miRNA-122-5p inhibition, we performed loss-of-function and gain-of-function experiments with an miRNA-122-5p inhibitor and mimic and quantified *BCL2* gene expression in cardiomyocytes and their corresponding EVs (Figure 7E and 7F). We found that inhibition of miR-122-5p augments *BCL2* expression, whereas overexpression suppresses *BCL2* in cardiomyocytes and their corresponding EVs. To investigate whether the stability of *BCL2* mRNA is regulated by miR-122-5p, we treated cells with actinomycin D to block transcription of *BCL2* before the inhibition of miR-122-5p. Consistent with the in silico prediction, these experiments indicated that miR-122-5p downregulates transcription of *BCL2* in a time-dependent manner (Figure 7G).

To determine the involvement of BCL2 as a mediator of miR-122-5p-dependent regulation in cell viability and apoptosis, we silenced *BCL2* and miR-122-5p. Reduced miR-122-5p expression prevented BCL2-dependent apoptosis and increased the viability of cardiomyocytes, which suggests that inhibition of miR-122-5p and inhibition of *BCL2* have opposing functions (Figure S5C and S5D). Last, we investigated BCL2 protein expression by using immunofluorescence and immunoblotting to examine the impact of miR-122-5p inhibition on the translation of BCL2 (Figure 7H and Figure S5E) and found that miR-122-5p is a crucial regulator of *BCL2*, both transcriptionally and posttranscriptionally. Together, these data confirmed that regulation of *BCL2* by miR-122-5p triggers apoptosis of cardiomyocytes and that the level of expression of miR-122-5p is important for this mechanism.

DISCUSSION

The data from the present study demonstrate a newly identified role for EV-miRNAs associated with improvements in heart function after TAVR in patients with AVS with reduced LVEF. MiR-122-5p expression is correlated with LVEF alterations and was found to have a predictive and prognostic value for cardiac recovery. On the basis of our clinical findings, we further explored the functional role of miR-122-5p. Of note, the level of miR-122-5p was higher in cardiac cells isolated from AVS mice compared with control mice and was released into the circulation in a vesicle-bound form. We found a 4- to 8-nucleotide sequence (*AAMRUGCU*) present in miR-122-5p that is known to regulate the packaging of miRNAs into large EVs by a multifunctional RBP, hnRNPU, acting as a direct interaction partner. This interaction between miR-122-5p and hnRNPU may stabilize the miRNAs by direct binding, thus protecting them from degradation by nucleases. Further investigation demonstrated that EV-mediated shuttling of miR-122-5p plays a critical role in intercellular communication to regulate the viability and apoptosis of target cardiomyocytes. This effect is regulated by BCL2, a known mediator of viability and apoptosis.

In cardiovascular research, a wide range of studies have reported dynamic alterations in the expression of circulating miRNAs under different cardiovascular conditions, leading to interest in circulating miRNAs for clinical diagnosis and prognosis.[38–46] In patients with AVS, some have impaired heart function, as evidenced by a low LVEF, indicating the severity of the pathological changes of the failing heart and the urgency for TAVR treatment. In many patients, TAVR relieves symptoms and reduces the burden on the weakened heart, which often leads to an improvement of LV function.[47–52] However, some patients do not see an improvement in LVEF after TAVR, and we identified a group of patients with impaired cardiac performance that correlated with reduced LV function, even with flawless TAVR treatment. To the best of our knowledge, the current study is the first to compare miRNA levels of patients with different responses to TAVR after a relatively long-term follow-up (6 months). By using an unbiased approach, we discovered that miR-122-5p and miR-1274a (miR-122-5p, *P*=0.005; miR-1274a, *P*=0.04; Figure S1B and Table 2) could be biomarkers for the recovery of LV function after TAVR. Multivariable logistic regression analysis indicated that the change in miR-122-5p was significantly associated with LVEF improvement. Patients with robust LVEF

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improvement at 6 months have lower miR-122-5p levels after TAVR, and circulating miR-122-5p levels were increased in patients with no LVEF improvement. The correlation between changes in LVEF and changes in the level of miR-122-5p in plasma for patients at day 7 after TAVR indicates that an elevation of miR-122-5p levels could predict a worse cardiac outcome in patients with no or low LVEF improvement. Most important, several studies have demonstrated that the level of miR-122-5p is higher in patients with heart failure compared with healthy control subjects, in agreement with our data.[43,53] Along those lines, we found that patients with AVS with low LVEF have higher levels of miR-122-5p expression than those with higher LVEF, indicating that miR-122-5p could be a marker of impaired LV function. When comparing the robust-improvement group and the no-improvement group, we observed a significant difference in the transvalvular pressure gradient. Patients in the unfavorable outcome group were more likely to have a low pressure gradient, whereas patients with a high pressure gradient displayed a better outcome. One possible mechanism to explain this could be that under low LVEF conditions, high pressure gradients may reflect that the heart of patients with stenosis, although already suffering from pressure overload and deteriorating remodeling, still has enough pumping power to create the pressure needed to generate adequate blood flow through the narrowing valve orifice. Similarly, we found that patients with robust LVEF improvement also showed improvement in echocardiographic parameters (change in LV end-diastolic volume index and stroke volume index) at 6 months after TAVR, which demonstrates signs of emerging reverse remodeling.

One recent study suggested that there is higher expression of miR-122-5p in a rat model of postinfarction heart failure than in a healthy control.[53] Cardiomyocytes isolated from mice with atrial fibrillation expressed more miR-122 than control mice, correlating with LVEF. To investigate whether miR-122-5p also is involved in AVS-related heart failure, we isolated RNA from the LV tissue of our murine AVS model. We found that levels of miR-122-5p were higher in AVS mice with reduced LVEF than in the sham-operated mice. This result supports the hypothesis that miR-122-5p expression can be modulated in response to different pathological conditions leading to reduced LVEF. In addition, in patients with heart failure, miR-122-5p levels in the coronary sinus are significantly higher than in peripheral arterial blood, which indicates the release of miR-122-5p during heart failure.[53] Decreased fractional shortening of cardiomyocytes is characterized by cardiomyocyte death, thus resulting in the deterioration of cardiac function in the failing heart. Increasing evidence suggests that promoting the survival of cardiomyocytes and simultaneously inducing angiogenesis could be a promising therapeutic route for treating acute myocardial infarction.[53,54] Our data suggest that miR-122-5p regulates the antiapoptotic gene *BCL2* to promote the viability of cardiomyocytes and might also affect angiogenesis by regulating the transcription of a proangiogenic signaling pathway. MiR-122-5p may provide a link between gene expression in ECs and cardiomyocytes. We confirmed that the set of genes that are induced by miR-122-5p lead to antiapoptotic and prosurvival effects in cardiac cells. One of the key genes that is controlled by miR-122-5p is *BCL2*, which plays a central role in apoptosis and is known to be cardioprotective. However, this in vivo experiment was based on the AVS model without any direct regulation of miR-122-5p; therefore, we cannot prove that miR-122-5p is indispensable to the improvement of heart function in patients after TAVR. AVS is a highly complex pathological condition that cannot be comprehensively investigated in vitro. Taken together, our results provide the first evidence that miR-122-5p is an indicator of heart function in patients with low or no LVEF improvement after TAVR. In addition, these data suggest that there is the potential for therapeutic regulation of miR-122-5p in patients with deteriorating heart function attributable to AVS.

EVs play a crucial role in intercellular signaling and the regulation of cellular function, within the context of AVS and heart failure.[38–46] EVs can transfer proteins, cytokines, mRNAs, and noncoding RNAs to target cells and influence their behavior. Thus, EVs are emerging as crucial regulators of the progression of cardiovascular disease. We performed vesicle-RNA degradation assays and confirmed that miR-122-5p is packaged mainly into large EVs.[55] Several recent studies suggested that RBPs are capable of sorting of ncRNAs into different EV populations, although most of these studies were focused primarily on small EVs.[20–22,31,35,36,54] We investigated and characterized large EVs and found high expression of certain hnRNPs in the isolated large EVs, which indicates that hnRNPs may also be involved in miRNA sorting into large EVs.

Multiple studies have shown that hnRNPU is a multifunctional, nuclear-enriched RBP (also present in the cytosol) that binds to various mRNAs and miRNAs, regulates transcription/translation of genes, and controls shuttling of miRNAs into membranous vesicles by sequence-specifically binding to mRNA/miRNAs.[21,35,36,54] Such a binding motif is found within miR-122-5p (and many other miRNAs); it contains a 4- to 8-nucleotide-long "exo" motif, which has been reported to play a vital role in the sorting of similar miRNAs into small EVs (exosomes). Therefore, the involvement of hnRNPU, may be particularly relevant for exporting miR-122-5p into large EVs. Loss of hnRNPU led to a decrease in the expression of miR-122-5p in ECs and their corresponding EVs. Because hnRNPU binds to a number of miRNAs, including miR-122-5p, a trend toward decreased export of miR-122-5p is predictable. Our in silico motif-enrichment data predicted that the extent of regulation of miRNA

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export correlates with the predicted binding to hnRNPU and that the binding-sequence motif *AAMRUGCU* is present in miR-122-5p (Figure S4D).[21,31] Together, our data suggest that the miR-122-5p sorting mechanism may be regulated exclusively by hnRNPU.

Given that EV-mediated cell-cell transfer of miRNAs is crucial in different cardiovascular diseases, EV-mediated intercellular communication might also be important in patients with AVS with low/no LVEF improvement. We observed a higher percentage of apoptotic cells in vitro after transfecting a miR-122-mimic or incubating with miR-122-5p–containing large vesicles. This agrees with several publications that demonstrate the proapoptotic effects of miR-122-5p.[40,55–64] We explored the mechanism by which miR-122-5p could have an effect on apoptosis in cardiomyocytes and revealed that miR-122-5p binds to the 3′ UTR of *BCL2* mRNA. To gain more insight into how miR-122-5p regulates the stability of *BCL2* mRNA, we treated cardiomyocytes with actinomycin D, which suppresses mRNA-transcription, and found that miR-122-5p negatively regulates *BCL2* transcription. Immunoblots clearly demonstrated an upregulation of BCL2 protein in miR-122-5p–downregulated cardiomyocytes, suggesting that miR-122-5p plays a pivotal role in *BCL2* mRNA/BCL2 protein stability. BCL2 is known to control apoptosis by regulating the permeability of the outer mitochondrial membrane, and expression of BCL2 is downregulated in acute myocardial infarction and other cardiovascular diseases in which cardiomyocyte death is a crucial factor. Our in vitro data suggest that EV–miR-122-5p can be shuttled to cardiomyocytes to exert a proapoptotic role by controlling *BCL2* transcription and translation.

However, our study has some limitations, which should be mentioned. (1) Because we obtained plasma samples only immediate to the procedure (day 1 before TAVR, day 1 after TAVR, and day 7 after TAVR), the miR-122-5p levels after a longer follow-up time (consistent with echocardiography parameters) are unknown, which could affect the accuracy of the association reported. (2) An additional animal model that could mimic the afterload reduction in the AVS mouse still needs to be established. It would be advantageous to understand the underlying mechanisms for no improvement in LVEF by developing a relevant murine disease model.

CONCLUSIONS

The data presented here highlight a novel role for EV-enriched miR-122-5p, which is to regulate apoptosis in cardiomyocytes, with implications for cardiac function in patients without LVEF improvement after TAVR.

ARTICLE INFORMATION

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Disclosures

None.

Supplemental Material

Methods

Tables S1–S4

Figures S1–S5

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